

Implementation of the PID controller in the S7-300

- PID controller block available in the standard STEP-7 instruction set (elementary capabilities)
- PID controller blocks available in the dedicated "Standard PID Control" extension package (extended capabilities)
- PID controller hardware modules (control of fast, complex processes)

„Standard PID Control”

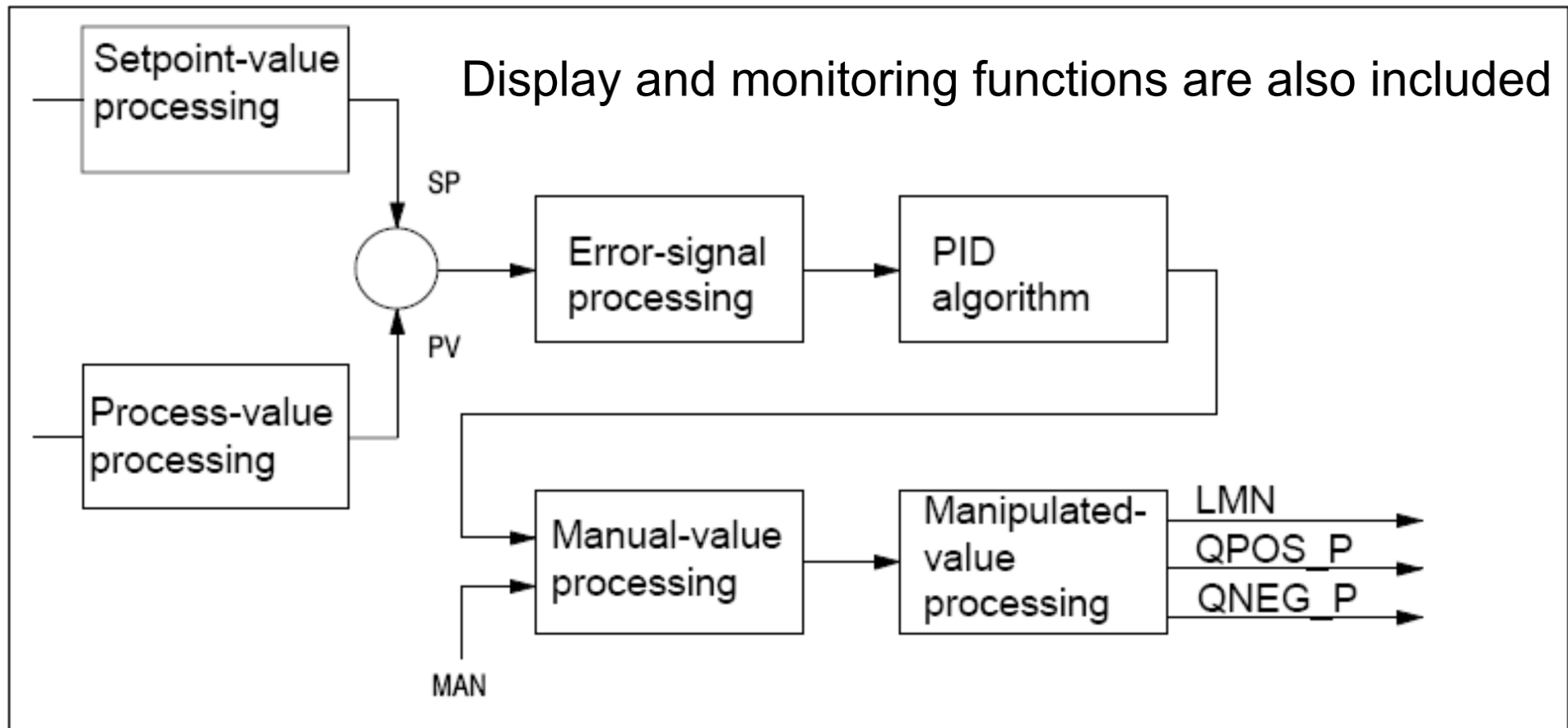
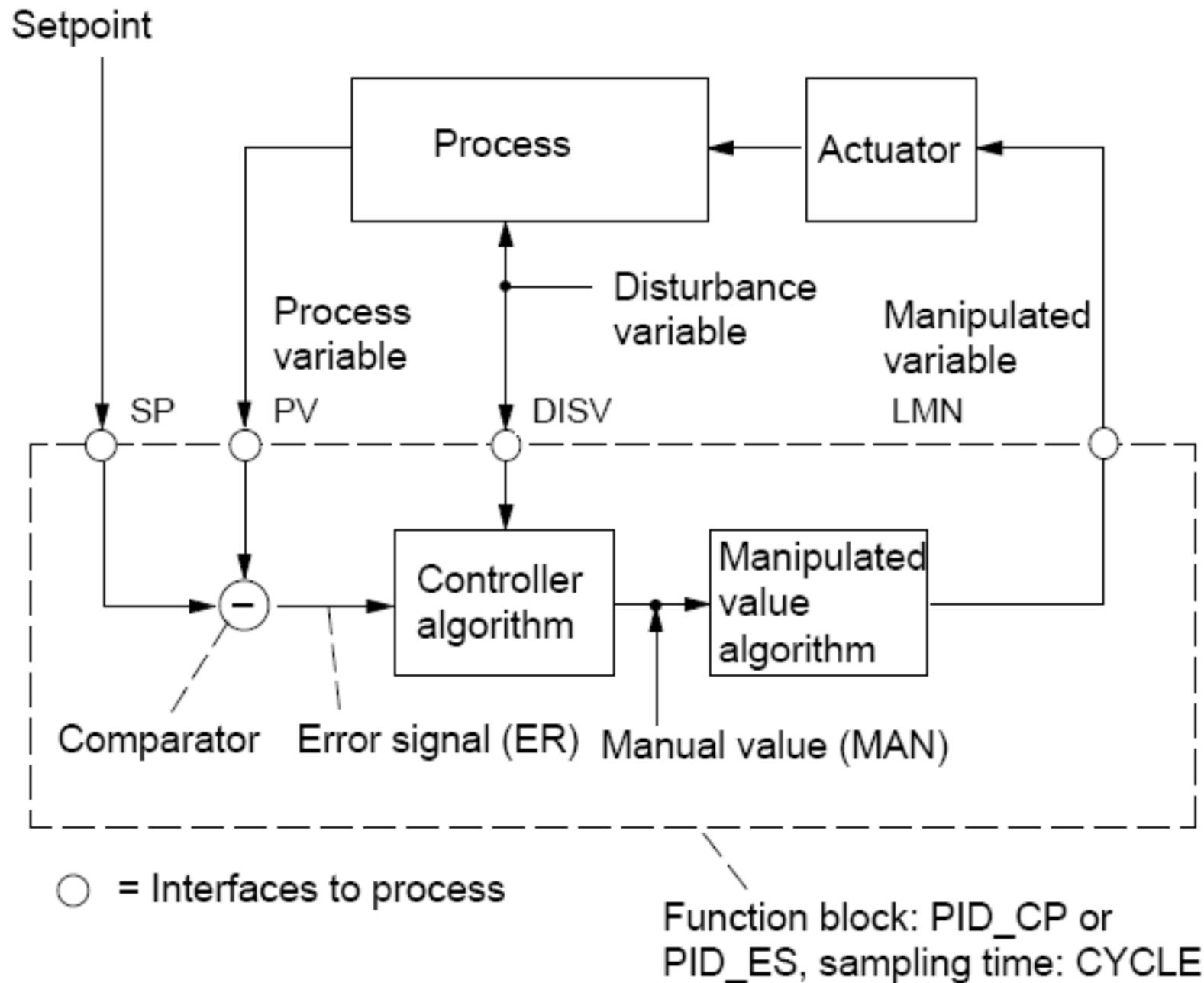


Figure 1-1 Function Overview of the Software Block "Continuous Controller"

PID block structure



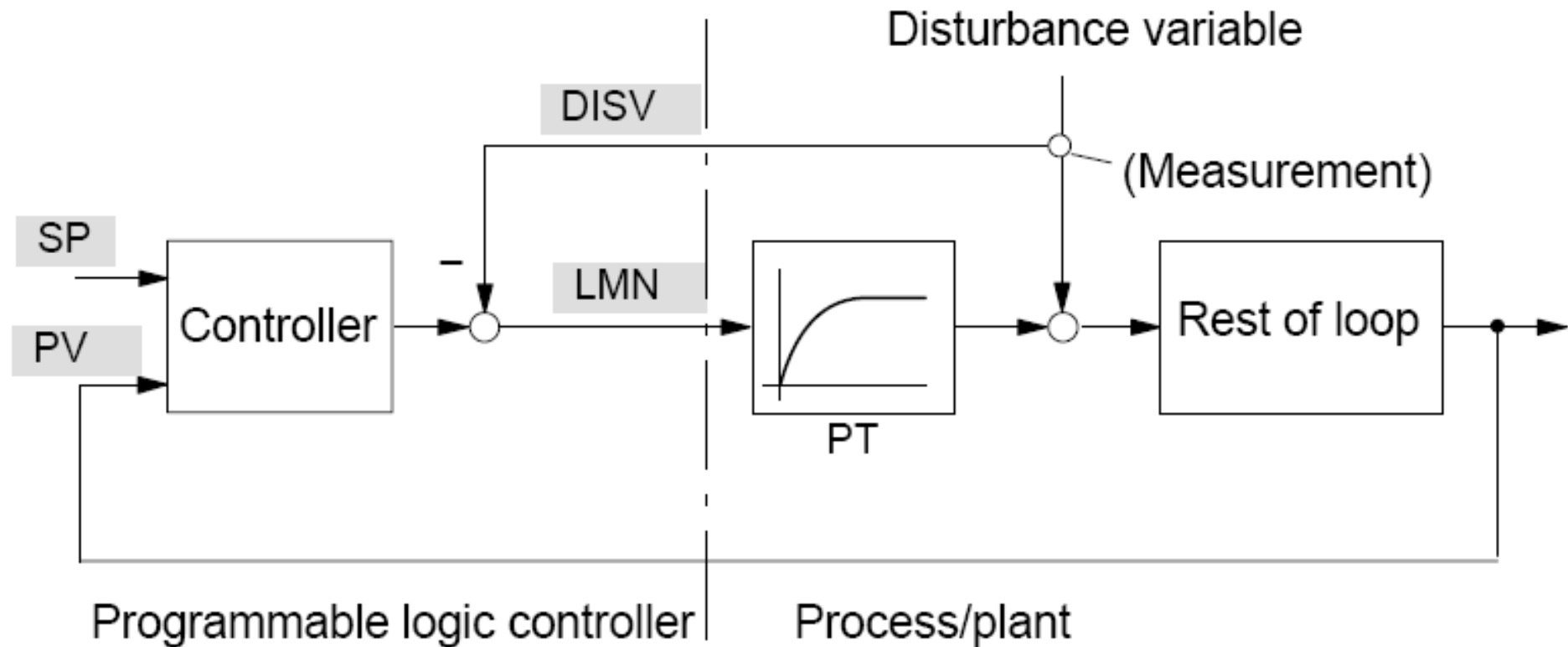
Using a standard PID block

- Fixed value control with P, PI, PD, PID
- Fixed value control with forward signal
- Cascade control
- Variable ratio control
- Mixing process control

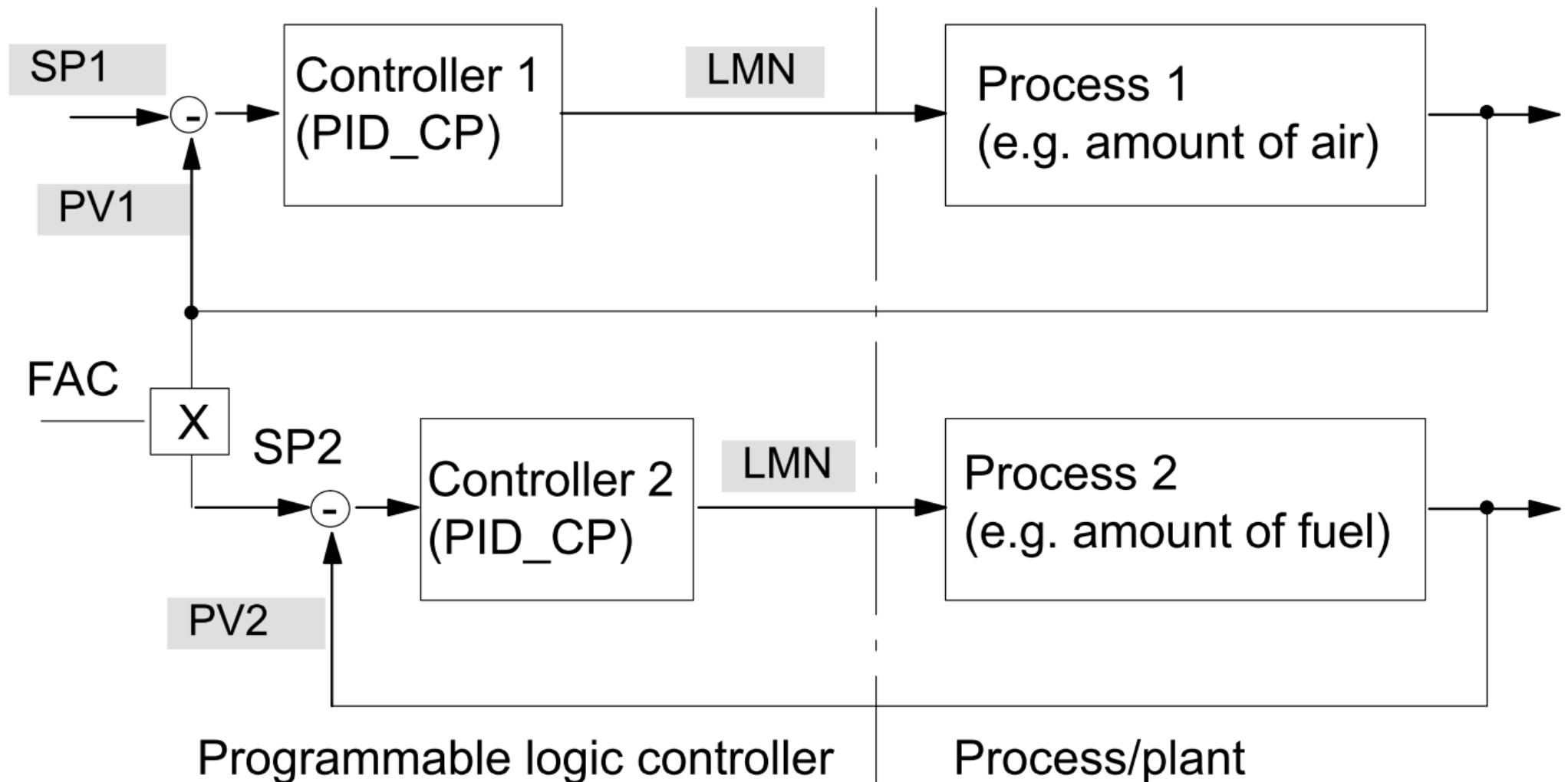
Standard PID block functions

- Limitation of signal values and rates of change of the setpoint and output signals
- Limitation of the influence of disturbances by filtering the deviation signal
- Additional delay of the measurement signal (inertial component) to reduce high frequency oscillations
- Linearisation of the quadratic function of the measurement signal (flow measurement by differential pressure sensor)
- Manual operation mode - setting the output value from other blocks
- Monitoring of two pairs of values: warning and alarm for measurement signals and deviation
- Possibility of using P and D action in the feedback path)

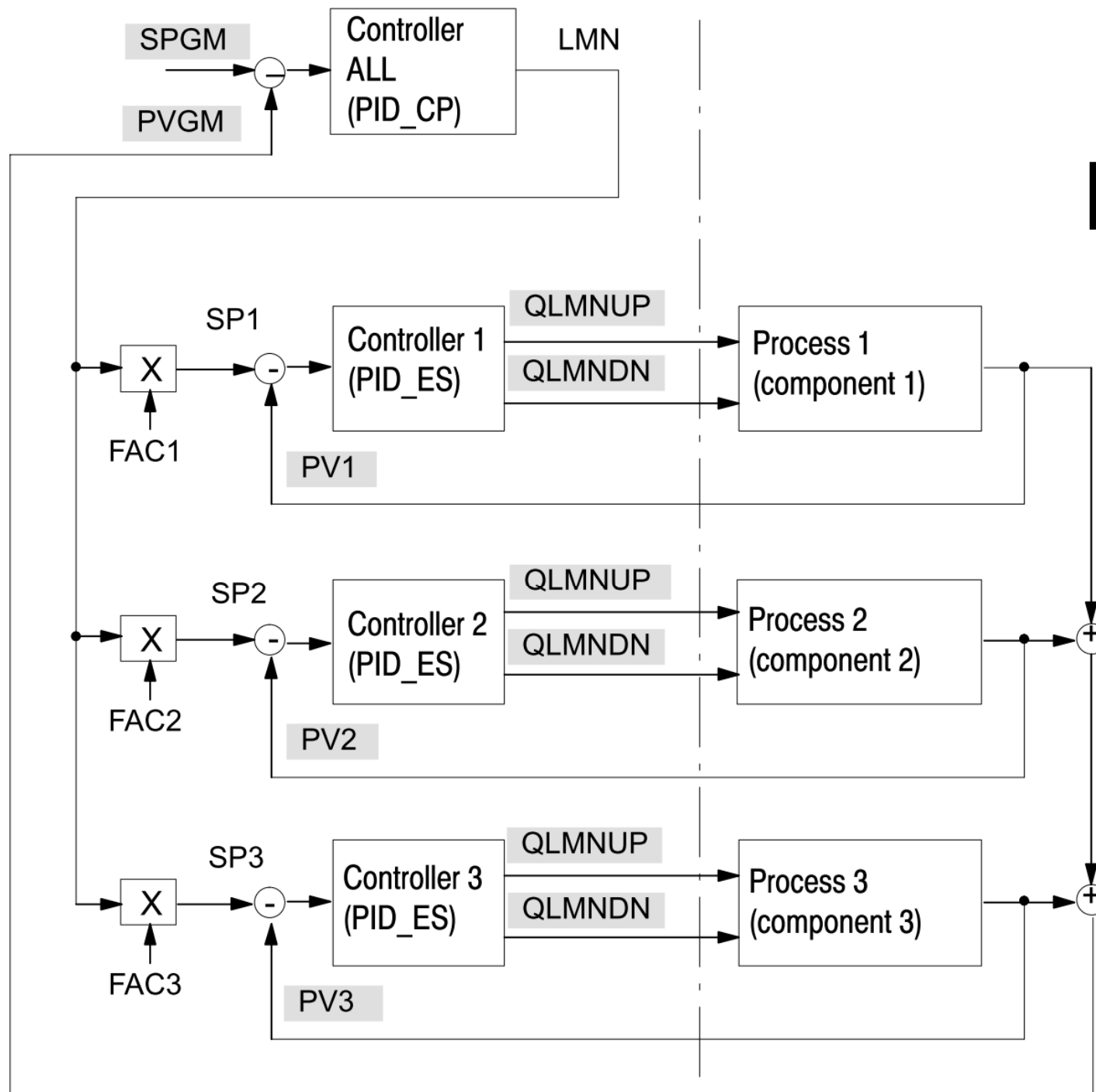
Compensation of the influence of measured disturbances



Ratio Control



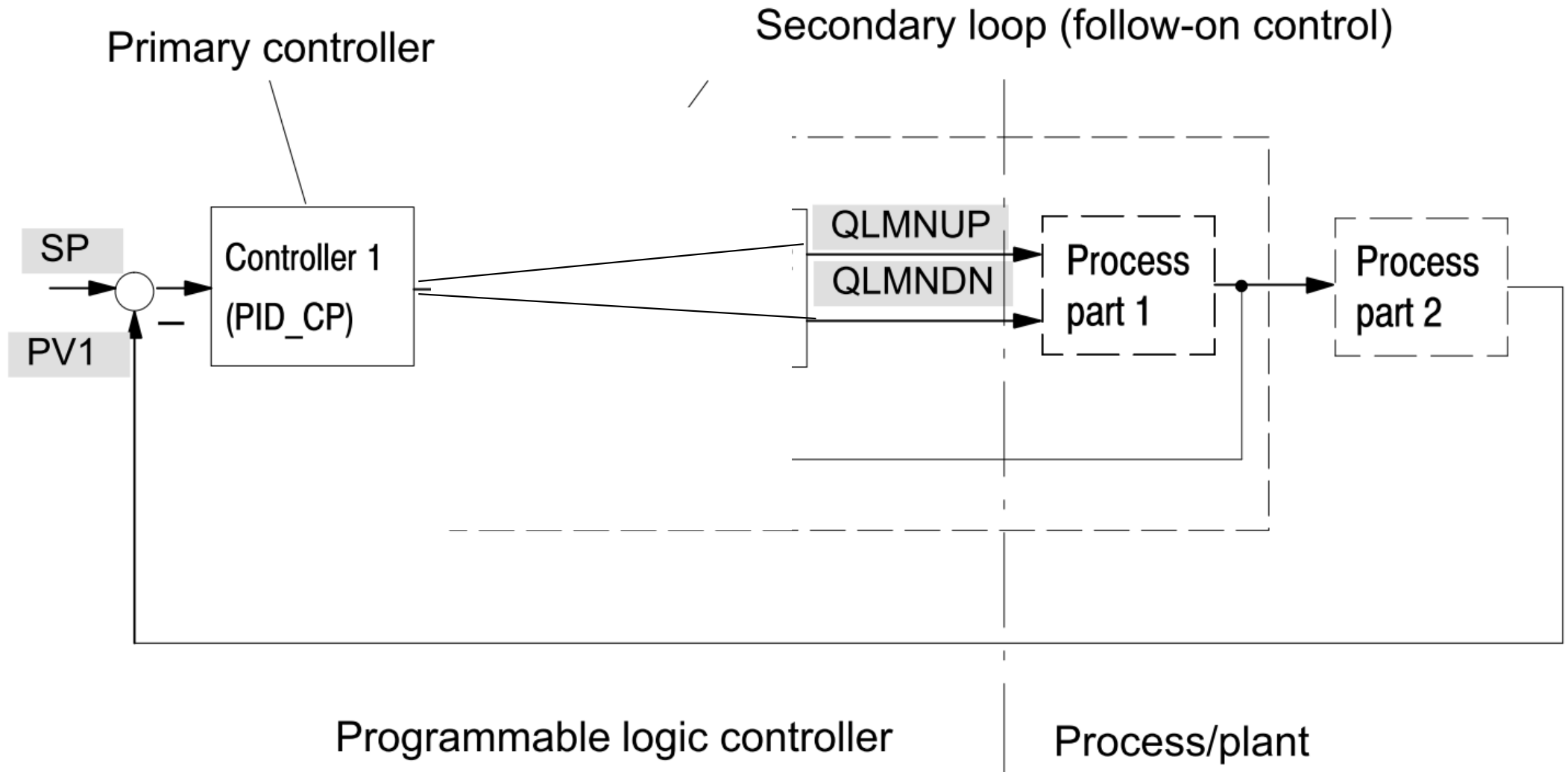
Mixing process control



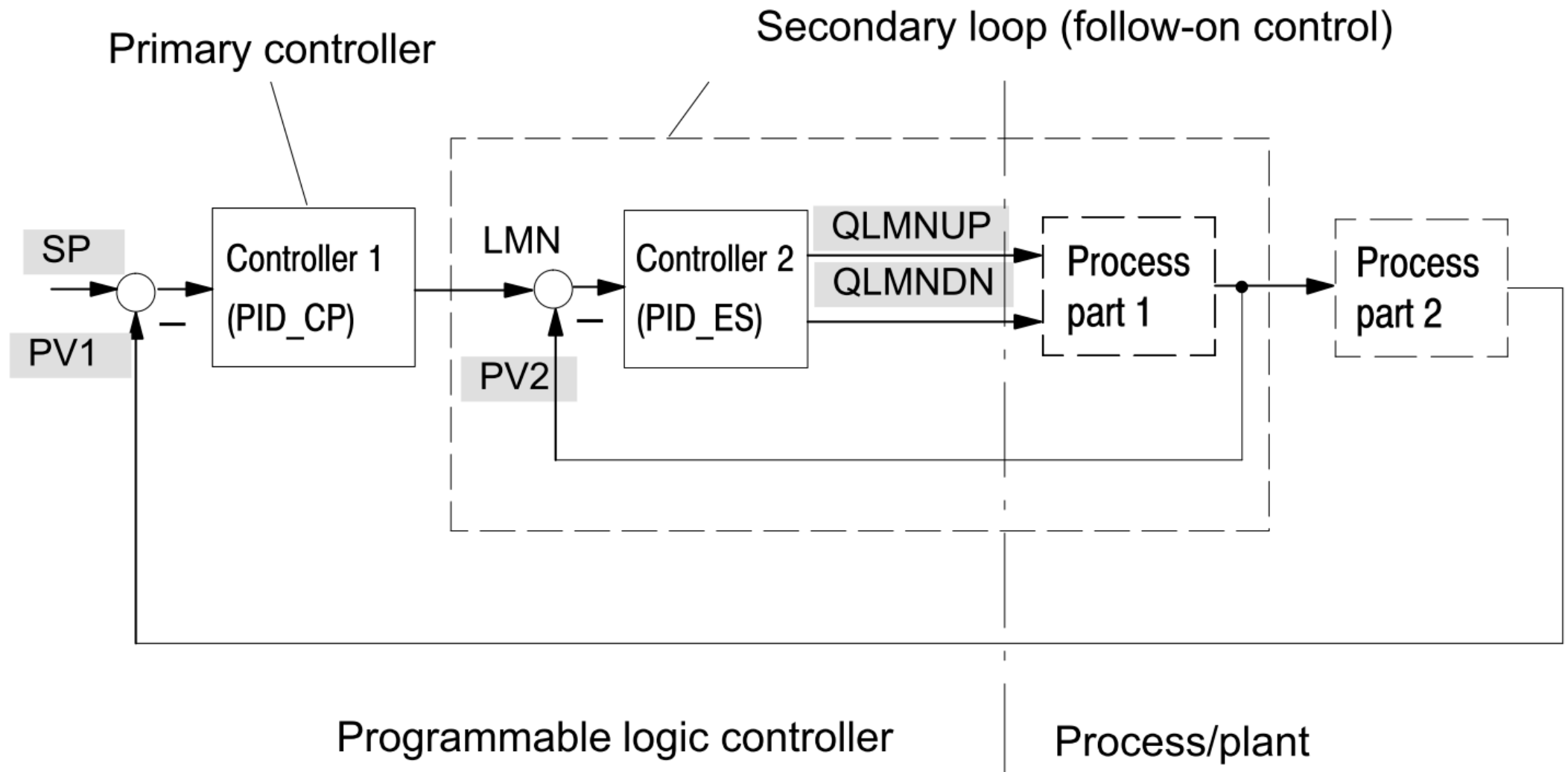
Programmable logic controller

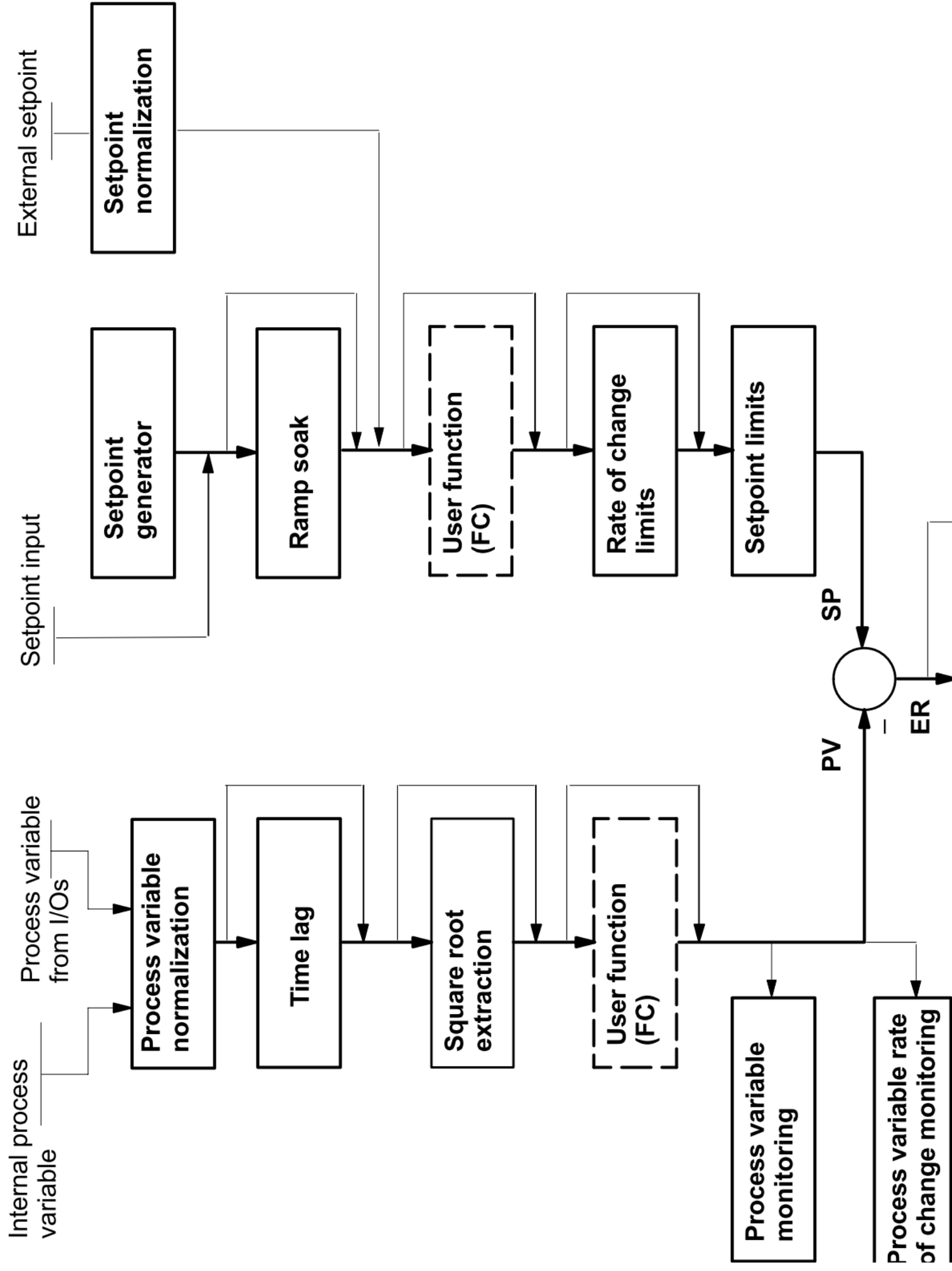
Process/plant

Cascade control

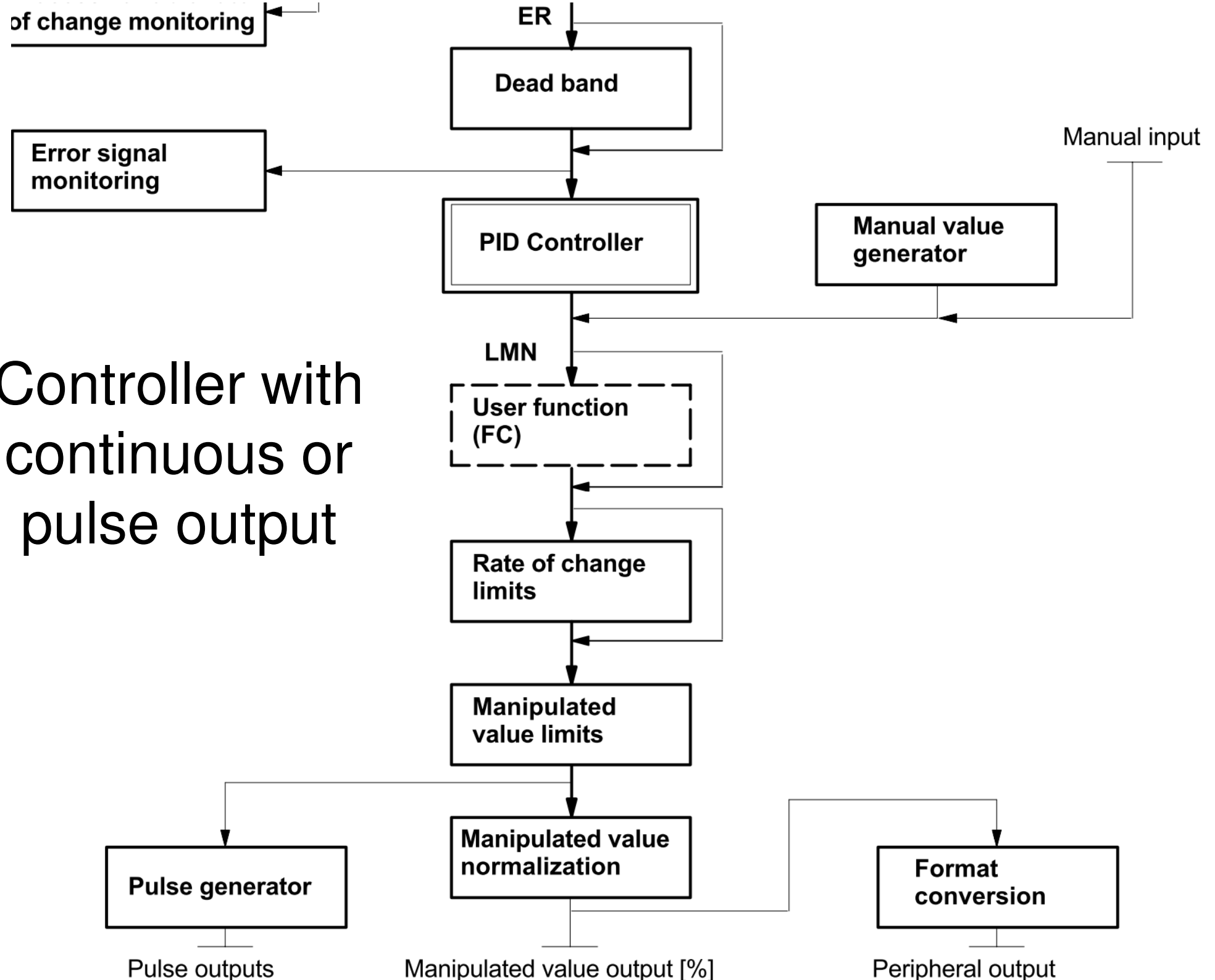


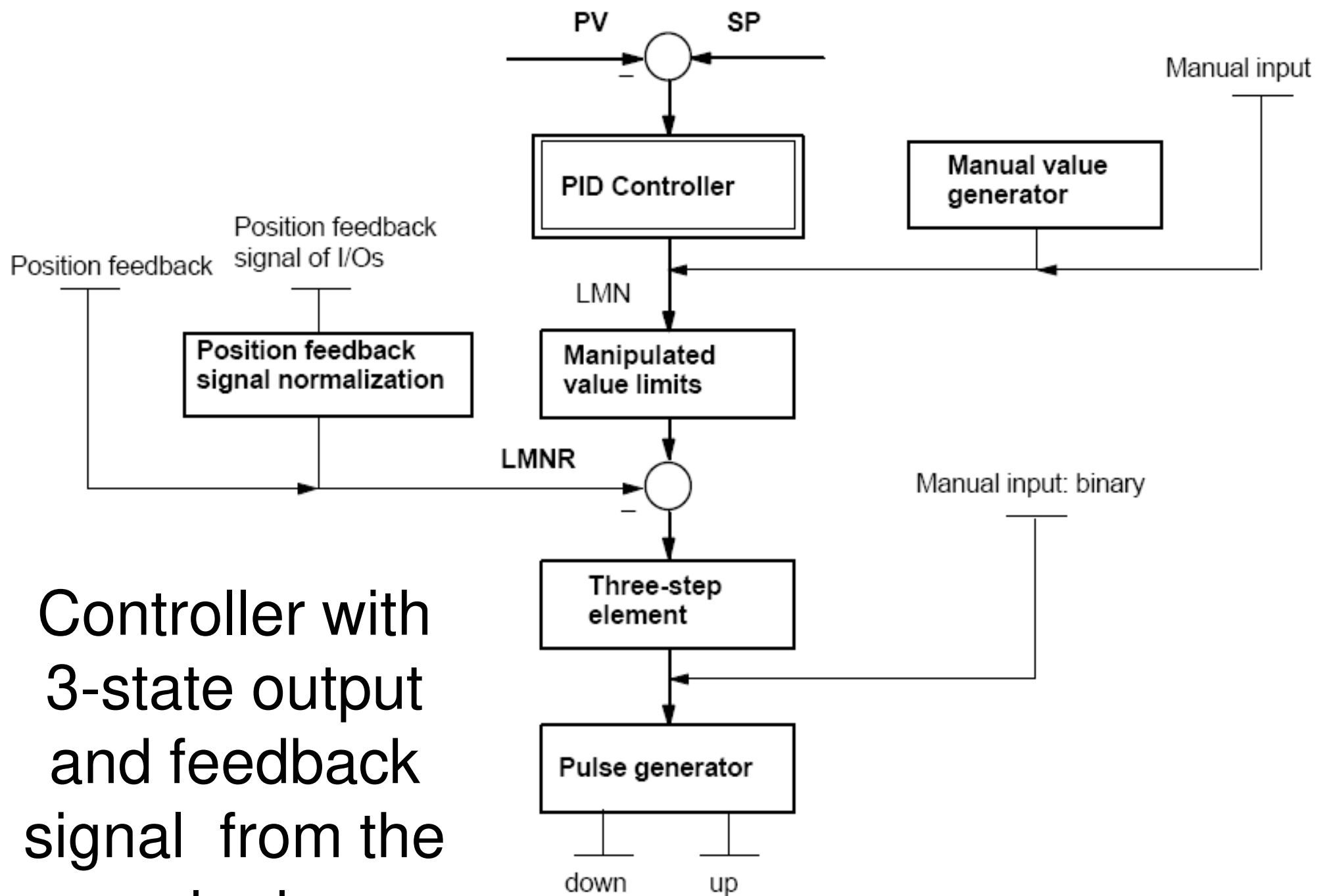
Cascade control



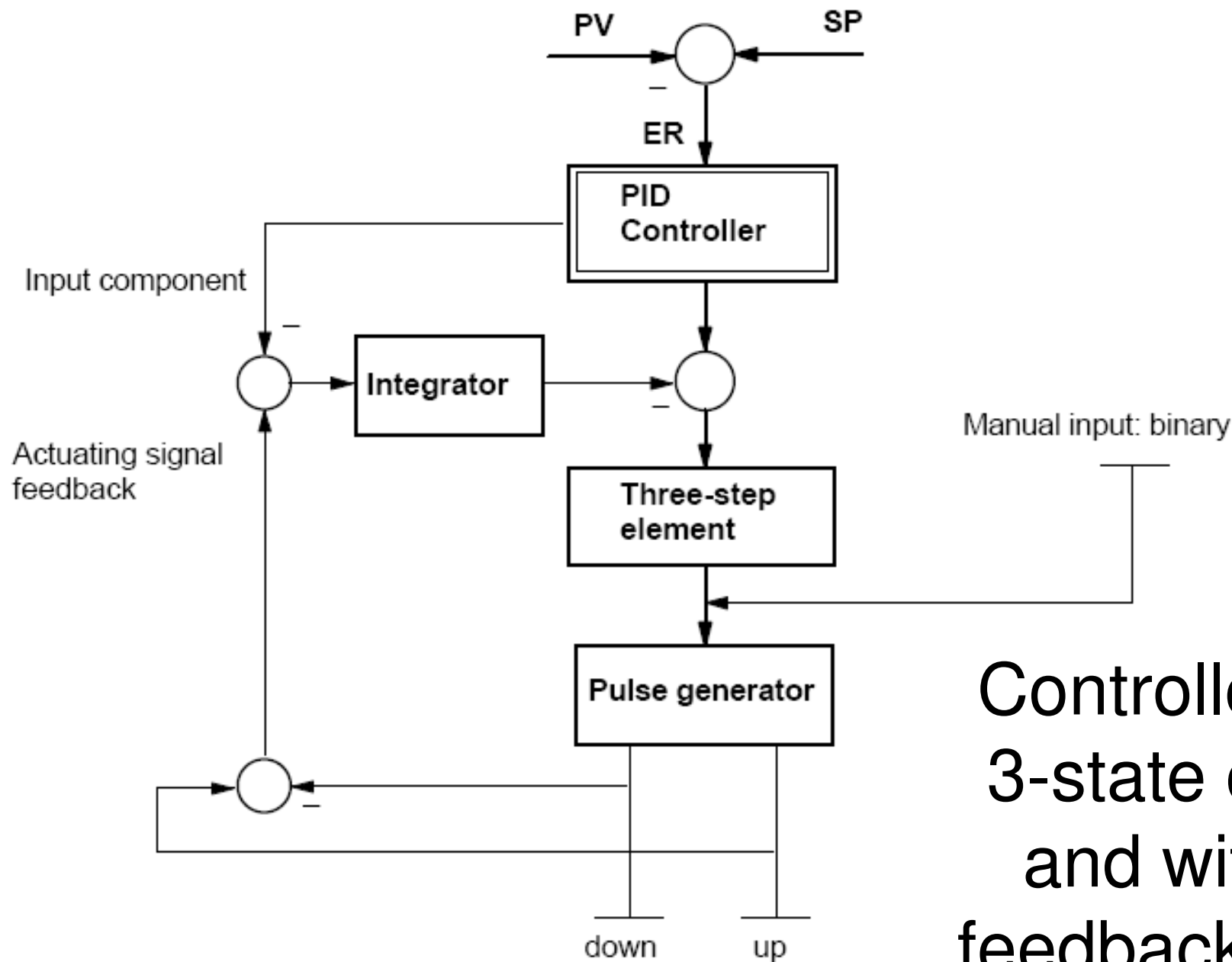


Controller with continuous or pulse output





Controller with
3-state output
and feedback
signal from the
actuator



Controller with
3-state output
and without
feedback signal
from the
actuator

DISV_SEL

DISV

P_SEL

I_SEL^{*})

D_SEL

GAIN

ER

ER Normalized

PID_OUTV

(Linear combination)

GAIN

ER

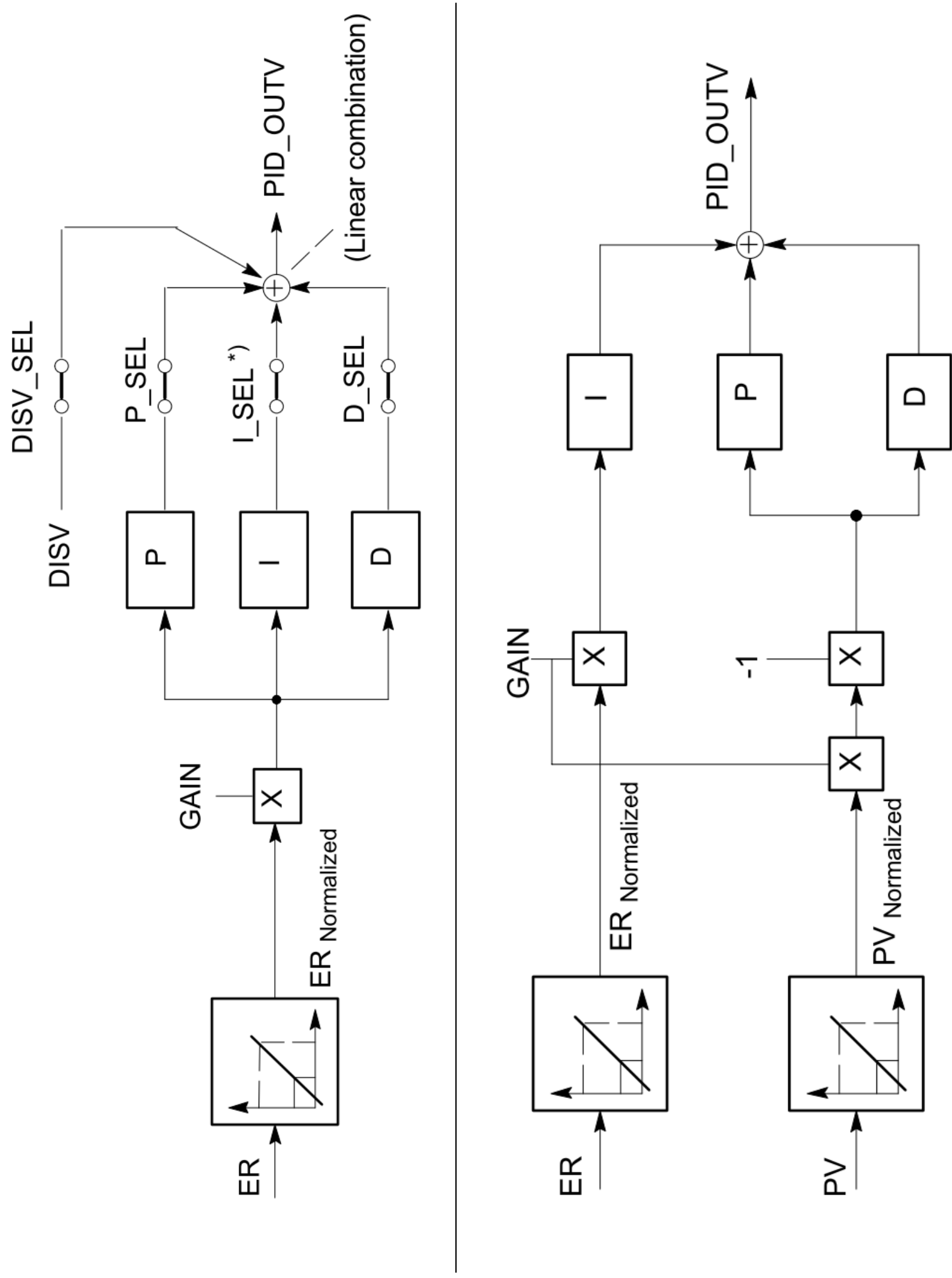
ER Normalized

PID_OUTV

-1

PV

PV Normalized

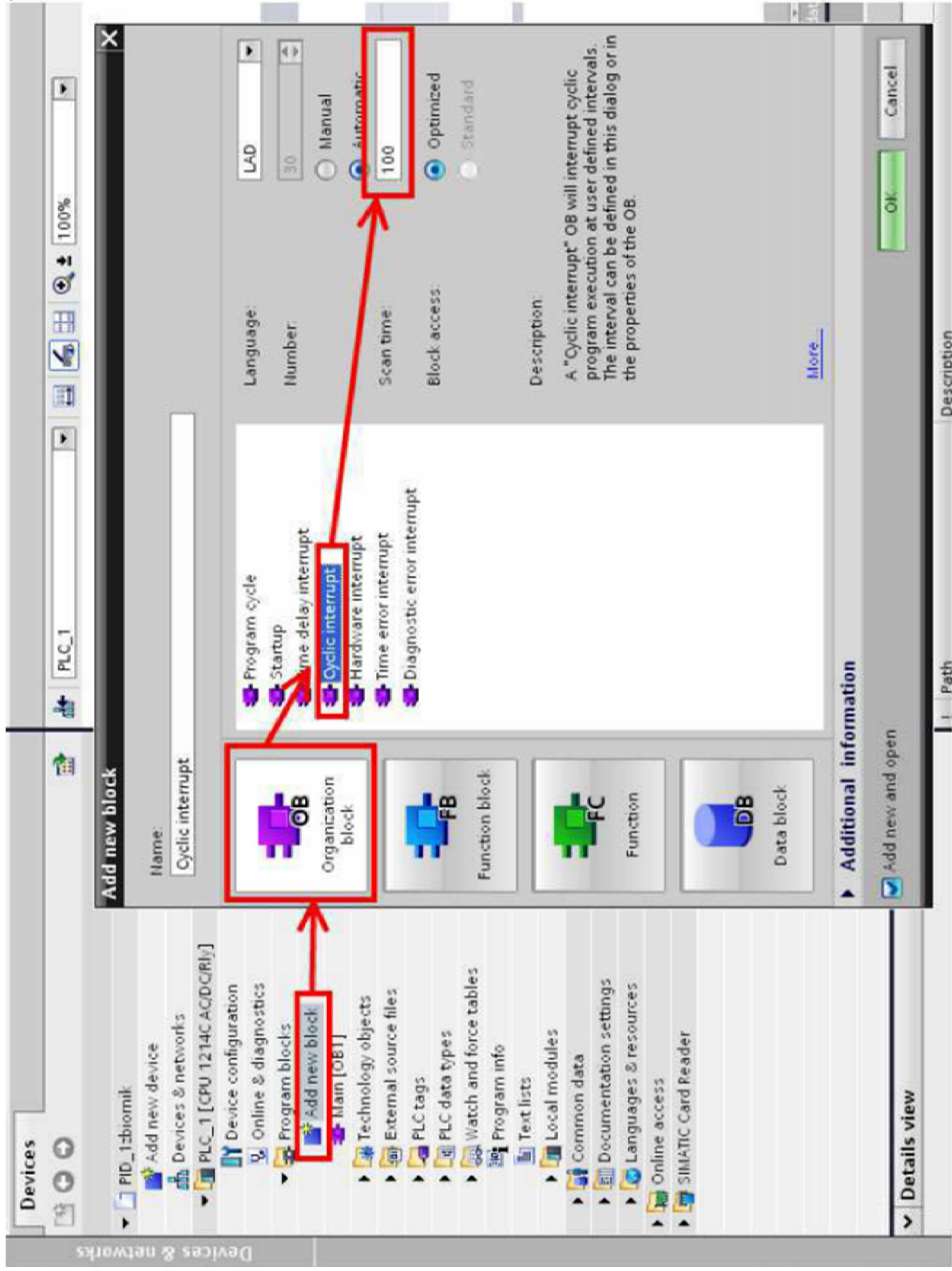


Types of PID blocks in the S7-1200

- **PID_Compact** - universal PID controller for process control
- **PID_3Step** - for process control with actuators controlled by on/off signals (e.g. valves with actuator)

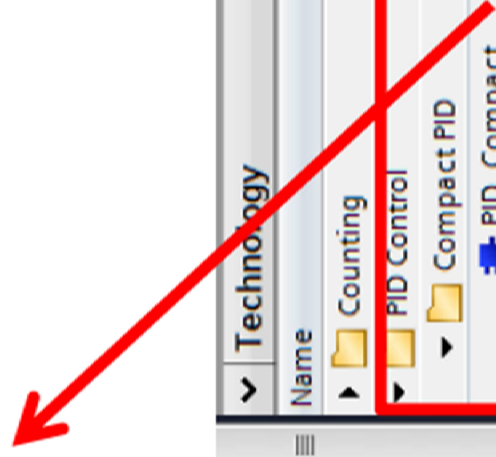
Controller parameters

- User-selected
- Selected through a self-tuning process (autotuning):
- "pretuning" mode (initial tuning): based on rise time and dead time,
- fine tuning mode (fine tuning): based on constant and limited oscillations (relay method).

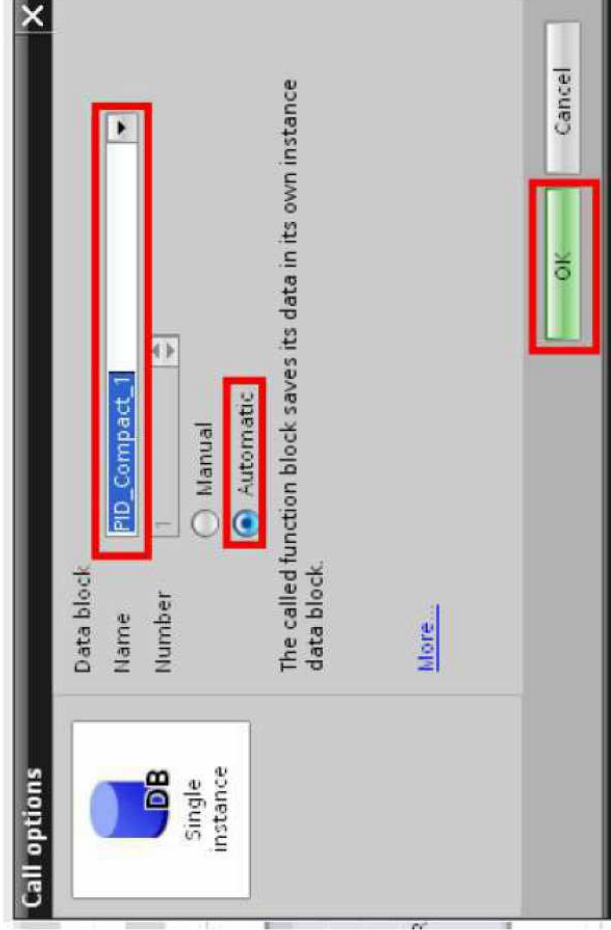


%DB1

PID_Compact_1



Technology			
Name	Description	Version	
Counting		V1.0	
PID Control			
Compact PID			
PID_Compact	Universal PID controller...	V1.2	
PID_3Step	PID controller with tuni...	V1.1	
Motion Control			



PID Compact

The PID controller uses the following formula to calculate the output value for the PID_Compact instruction.

$$y = K_p \left[(b \cdot w - x) + \frac{1}{T_i \cdot s} (w - x) + \frac{T_D \cdot s}{a \cdot T_D \cdot s + 1} (c \cdot w - x) \right]$$

y	Output value	x	Process value
w	Setpoint value	s	Laplace operator
K _p	Proportional gain (P component)	a	Derivative delay coefficient (D component)
T _i	Integral action time (I component)	b	Proportional action weighting (P component)
T _D	Derivative action time (D component)	c	Derivative action weighting (D component)

PID 3step

The PID controller uses the following formula to calculate the output value for the PID_3Step instruction.

$$\Delta y = K_p \cdot s \cdot \left[(b \cdot w - x) + \frac{1}{T_I \cdot s} (w - x) + \frac{T_D \cdot s}{a \cdot T_D \cdot s + 1} (c \cdot w - x) \right]$$

y Output value

w Setpoint value

K_p Proportional gain
(P component)

T_I Integral action time
(I component)

T_D Derivative action time
(D component)

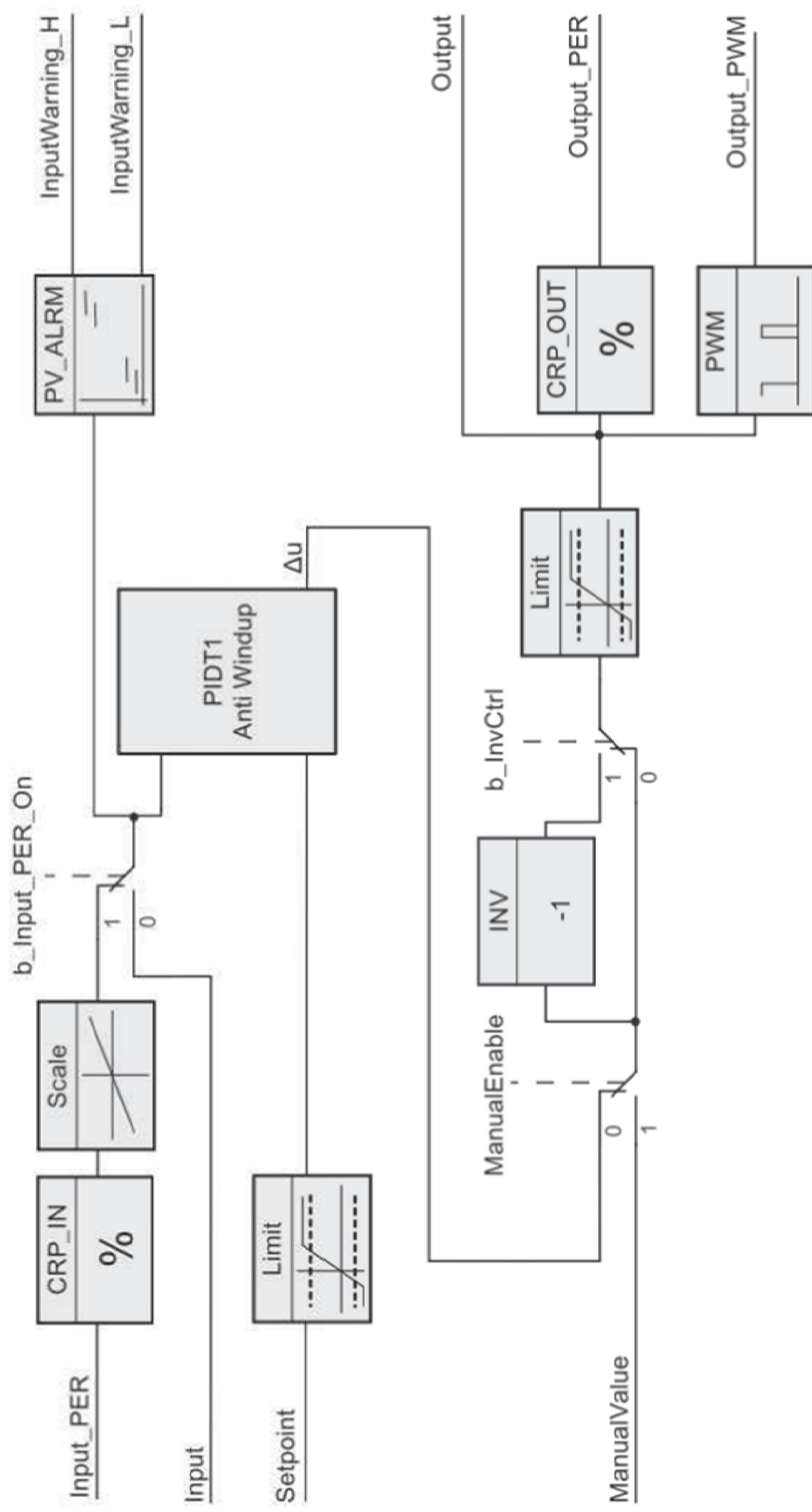
x Process value

s Laplace operator

a Derivative delay coefficient
(D component)

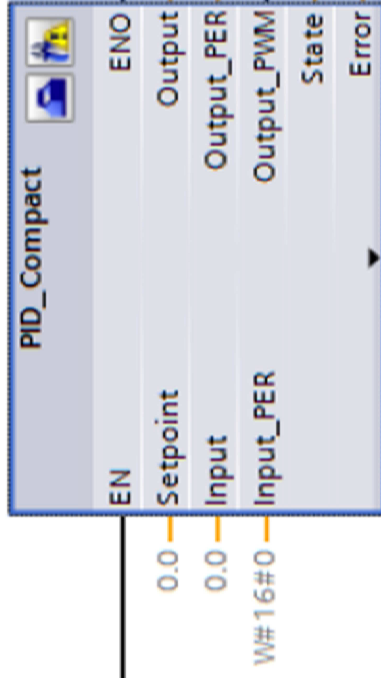
b Proportional action weighting
(P component)

c Derivative action weighting
(D component)



%DB1

"PID_Compact_1"



%DB1

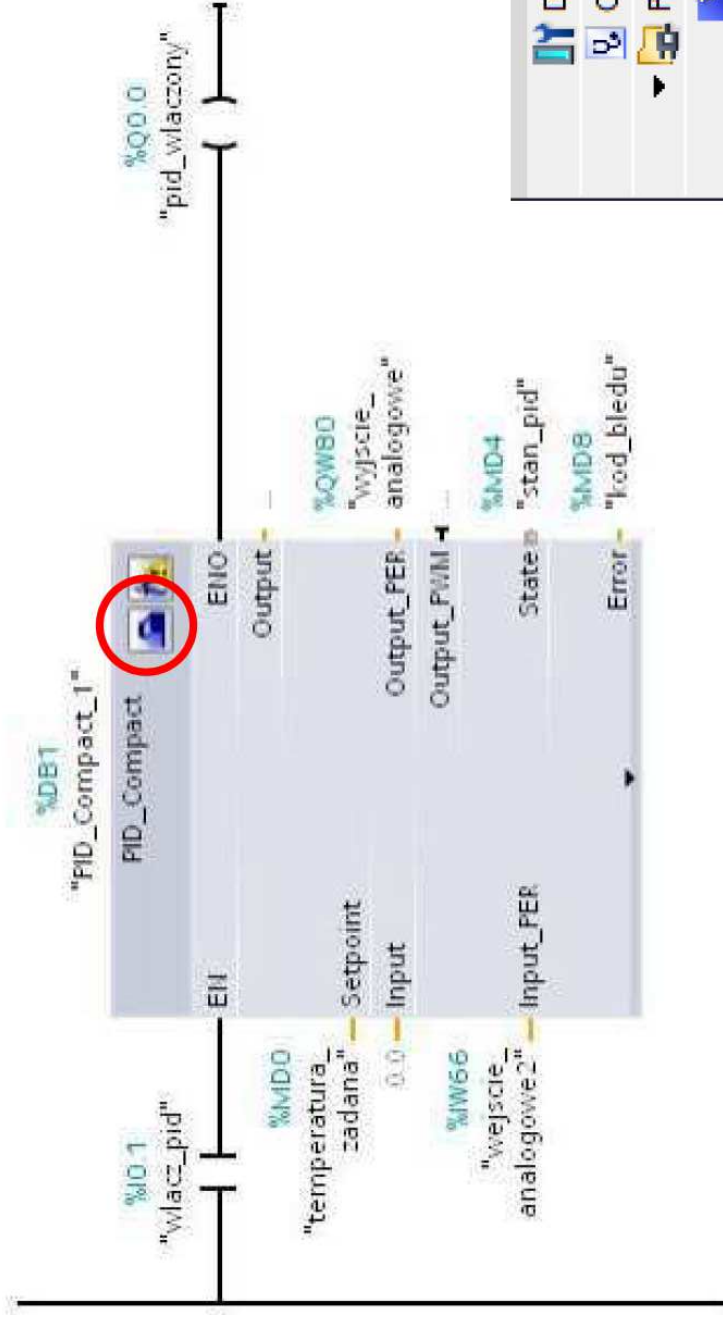
"PID_Compact_1"

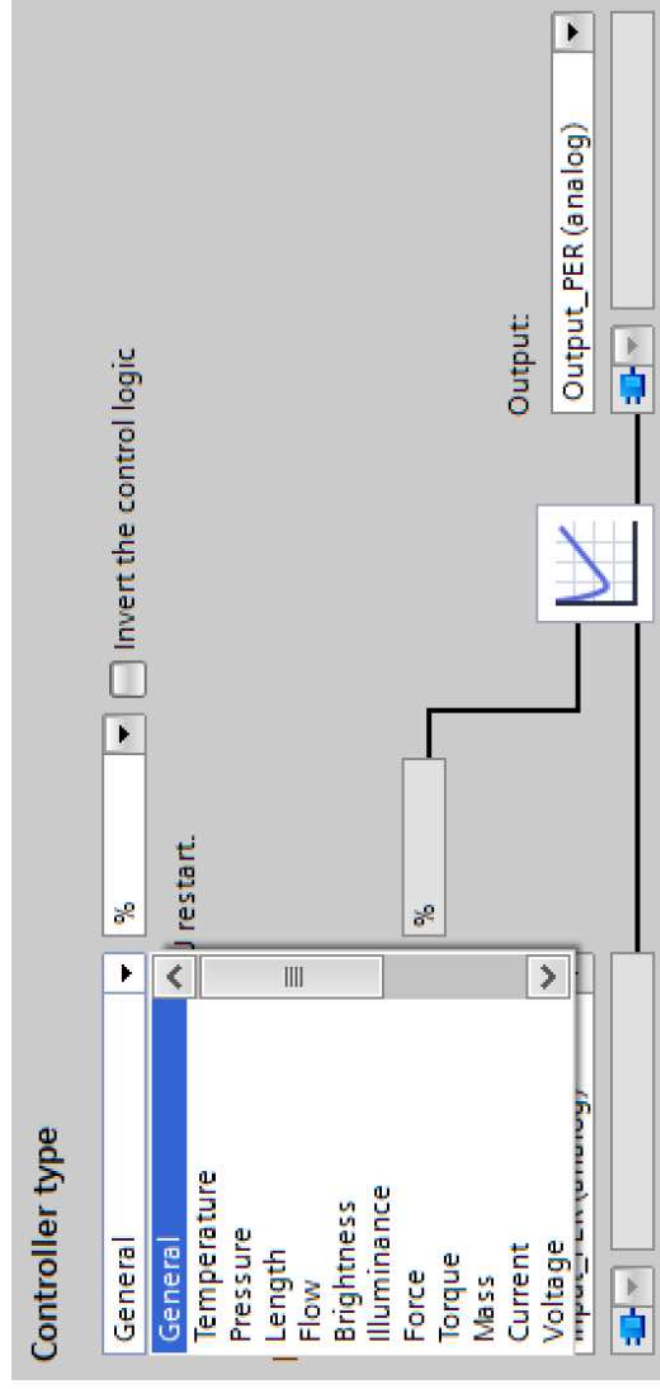
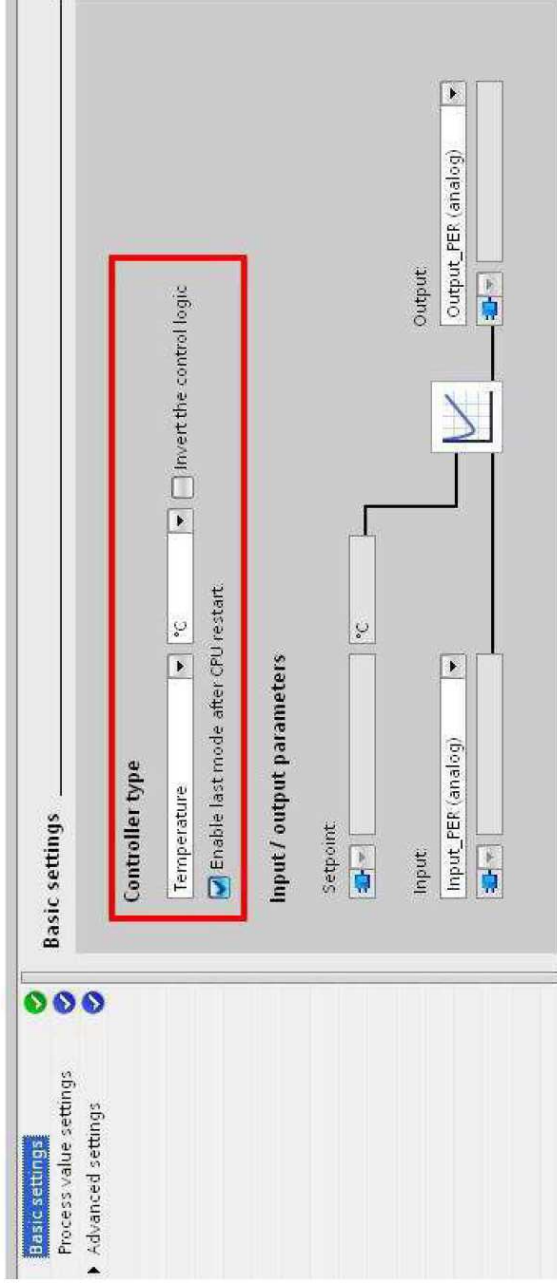


Parameter and type		Data type	Description
Setpoint	IN	Real	Setpoint of the PID controller in automatic mode. Default value: 0.0
Input	IN	Real	Process value. Default value: 0.0 You must also set sPid_Cmpt.b_Input_PER_On = FALSE.
Input_PER	IN	Word	Analog process value (optional). Default value: W#16#0 You must also set sPid_Cmpt.b_Input_PER_On = TRUE.
ManualEnable	IN	Bool	Enables or disables the manual operation mode. Default value: FALSE: - PID_Compact V1.0 and V1.2: When the CPU transitions to RUN, if the ManualEnable = TRUE, PID_Compact starts in manual mode. It is not necessary for a FALSE to TRUE transition to place the PID_Compact into manual mode. - PID_Compact V1.1: When the CPU transitions to RUN and the ManualEnable = TRUE, the PID Compact starts in the last state. A transition from TRUE to FALSE to TRUE is required to place the PID_Compact in manual mode.
ManualValue	IN	Real	Process value for manual operation. Default value: 0.0
Reset	IN	Bool	The Reset parameter restarts the controller. Default value: FALSE See the "Response to Reset" section below for PID_Compact V1.1 and V1.0 Reset response diagrams.
ScaledInput	OUT	Real	Scaled process value. Default value: 0.0
Output ¹	OUT	Real	Output value. Default value: 0.0
Output_PER ¹	OUT	Word	Analog output value. Default value: W#16#0
Output_PWM ¹	OUT	Bool	Output value for pulse width modulation. Default value: FALSE

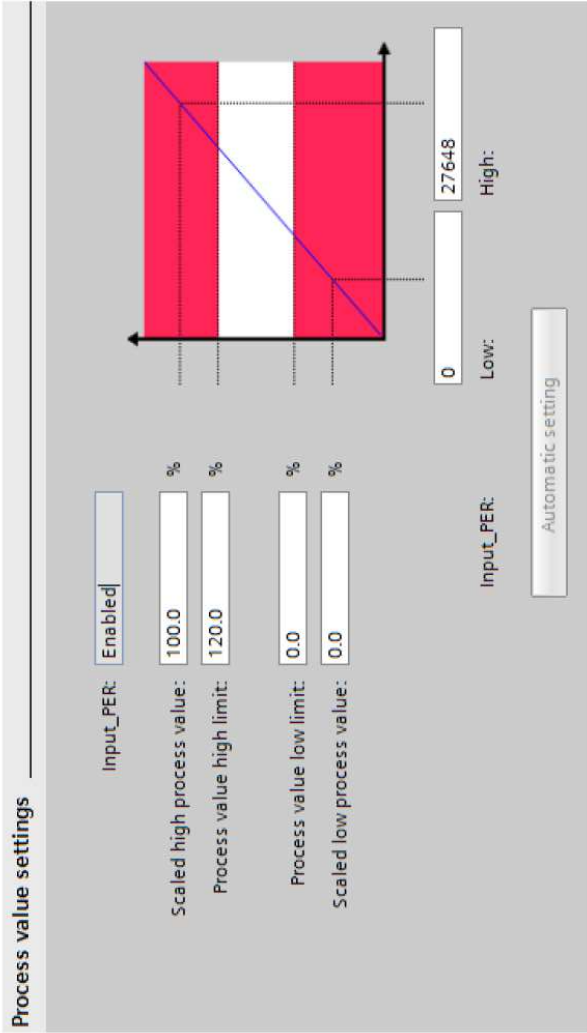
The outputs of the Output, Output_PER, and Output_PWM parameters can be used in parallel.

SetpointLimit_H	OUT	Bool	Setpoint high limit. Default value: FALSE If SetpointLimit_H = TRUE, the absolute upper limit of the setpoint is reached. Default value: FALSE
SetpointLimit_L	OUT	Bool	Setpoint low limit. Default value: FALSE If SetpointLimit_L = TRUE, the absolute lower limit of the setpoint is reached. Default value: FALSE
InputWarning_H	OUT	Bool	If InputWarning_H = TRUE, the process value reached or exceeded the upper warning limit. Default value: FALSE
InputWarning_L	OUT	Bool	If InputWarning_L = TRUE, the process value reached the lower warning limit. Default value: FALSE
State	OUT	Int	Current operating mode of the PID controller. Default value: 0 Use sRet.i_Mode to change the mode. • State = 0: Inactive • State = 1: Pretuning • State = 2: Manual fine tuning • State = 3: Automatic mode • State = 4: Manual mode
ErrorBits	OUT	DWord	The PID_Compact instruction ErrorBits parameters table (Page 354) defines the error messages. Default value: DW#16#0000 (no error)





Settings		Description
Basic	Controller type	Selects the engineering units.
	Invert the control logic	Allows selection of a reverse-acting PID loop. - If not selected, the PID loop is in direct-acting mode and the output of PID loop increases if input value < setpoint. - If selected, the output of the PID loop increases if the input value > setpoint.
	Enable last mode after CPU restart	Restarts the PID loop after it is reset or if an input limit has been exceeded and returned to the valid range.
	Input	Selects either the Input parameter or the Input_PER parameter (for analog) for the process value. Input_PER can come directly from an analog input module.
	Output	Selects either the Output parameter or the Output_PER parameter (for analog) for the output value. Output_PER can go directly to an analog output module.
Process value	Scales both the range and the limits for the process value. If the process value goes below the low limit or above the high limit, the PID loop goes to inactive mode and sets the output value to 0. To use Input_PER, you must scale the analog process value (input value).	



Basic settings

Process value settings

Advanced settings

Process value monitoring

PWM limits

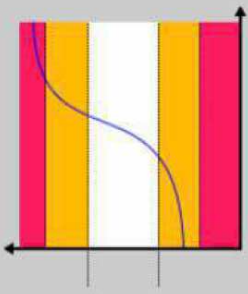
Output value limits

PID Parameters

Process value monitoring

Warning high limit: 3.402822E+38 %

Warning low limit: -3.402822E+38 %



PWM limits

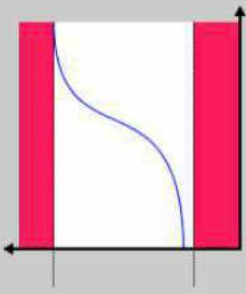
Minimum ON time: 0.0 s

Minimum OFF time: 0.0 s

Output value limits

Output value high limit: 100.0 %

Output value low limit: 0.0 %



Basic settings

Process value settings

Advanced settings

Process value monitoring

PWM limits

Output value limits

PID Parameters

PID Parameters

Enable manual entry

Proportional gain: 1.0

Integral action time: 20.0 s

Derivative action time: 0.0 s

Derivative delay coefficient: 0.0

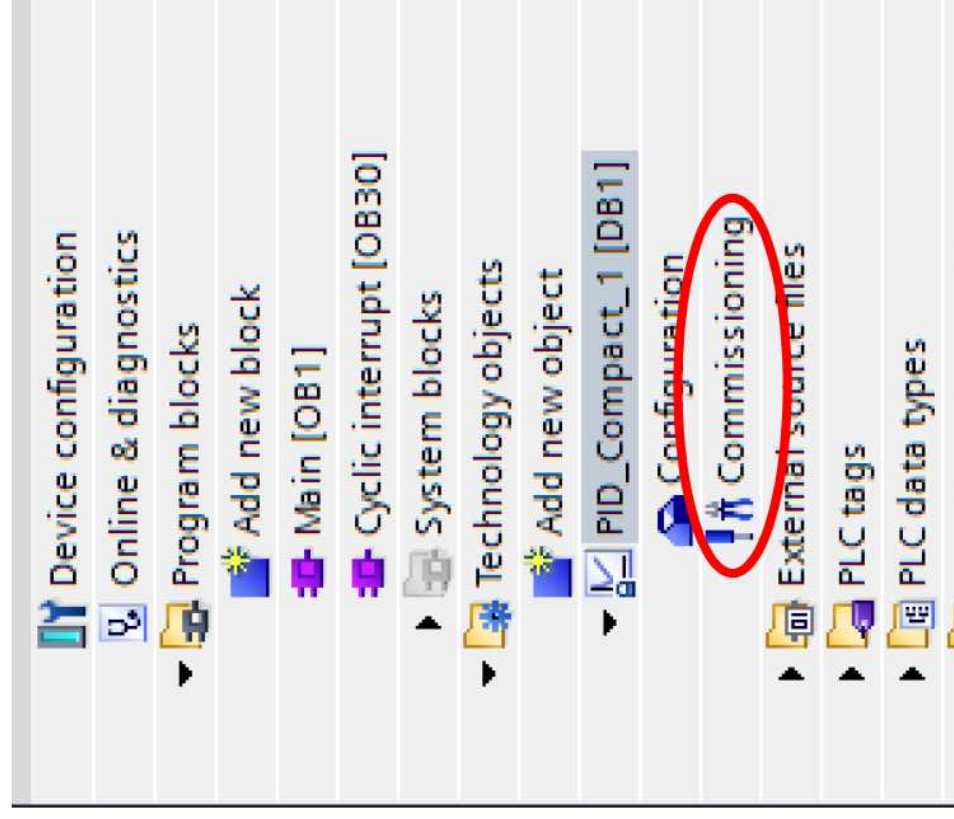
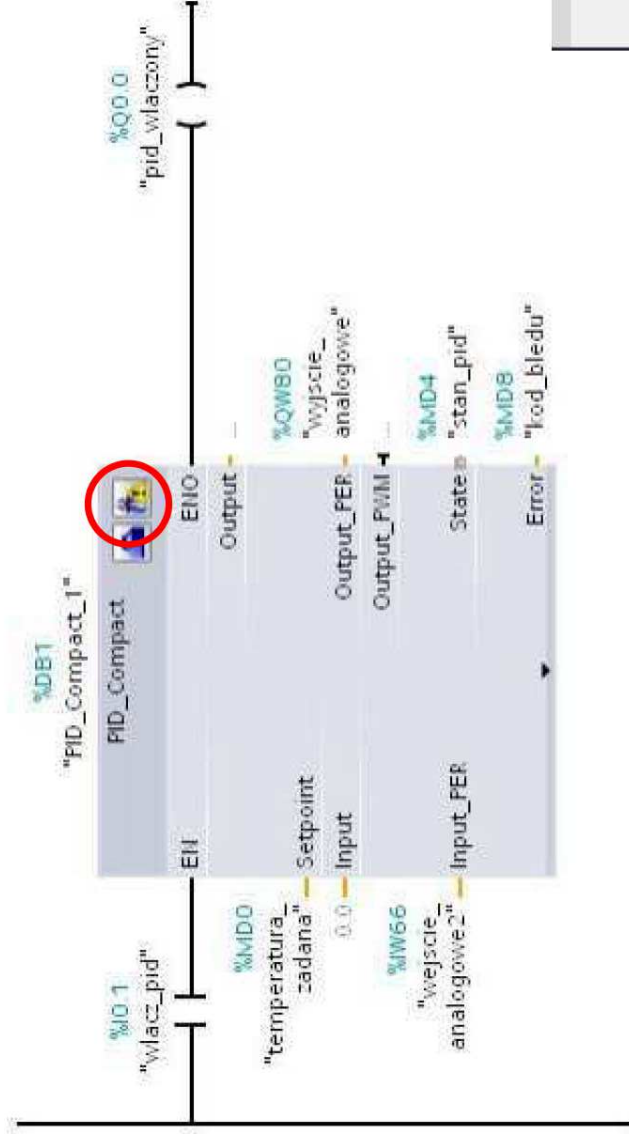
Proportional action weighting: 0.0

Derivative action weighting: 0.0

Sampling time of PID algorithm: 1.0 s

Tuning rule

Controller structure: PID



Tuning

Measurement:

Sampling time: 0.3 s



Tuning mode:

Pretuning



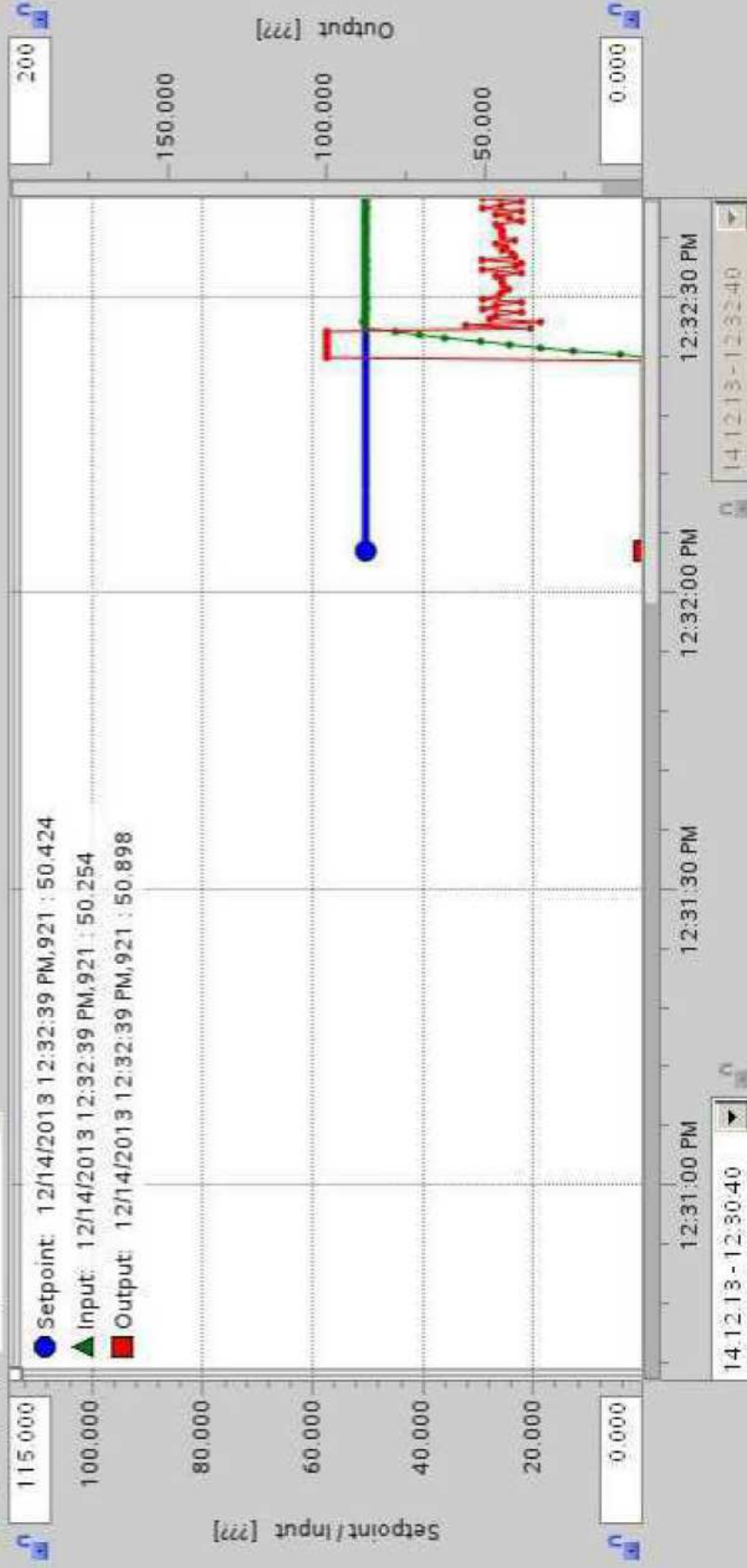
Strip



● Setpoint: 12/14/2013 12:32:39 PM, 921 : 50.424

▲ Input: 12/14/2013 12:32:39 PM, 921 : 50.254

■ Output: 12/14/2013 12:32:39 PM, 921 : 50.898



Tuning

Measurement:

Sampling time: 0.3 s



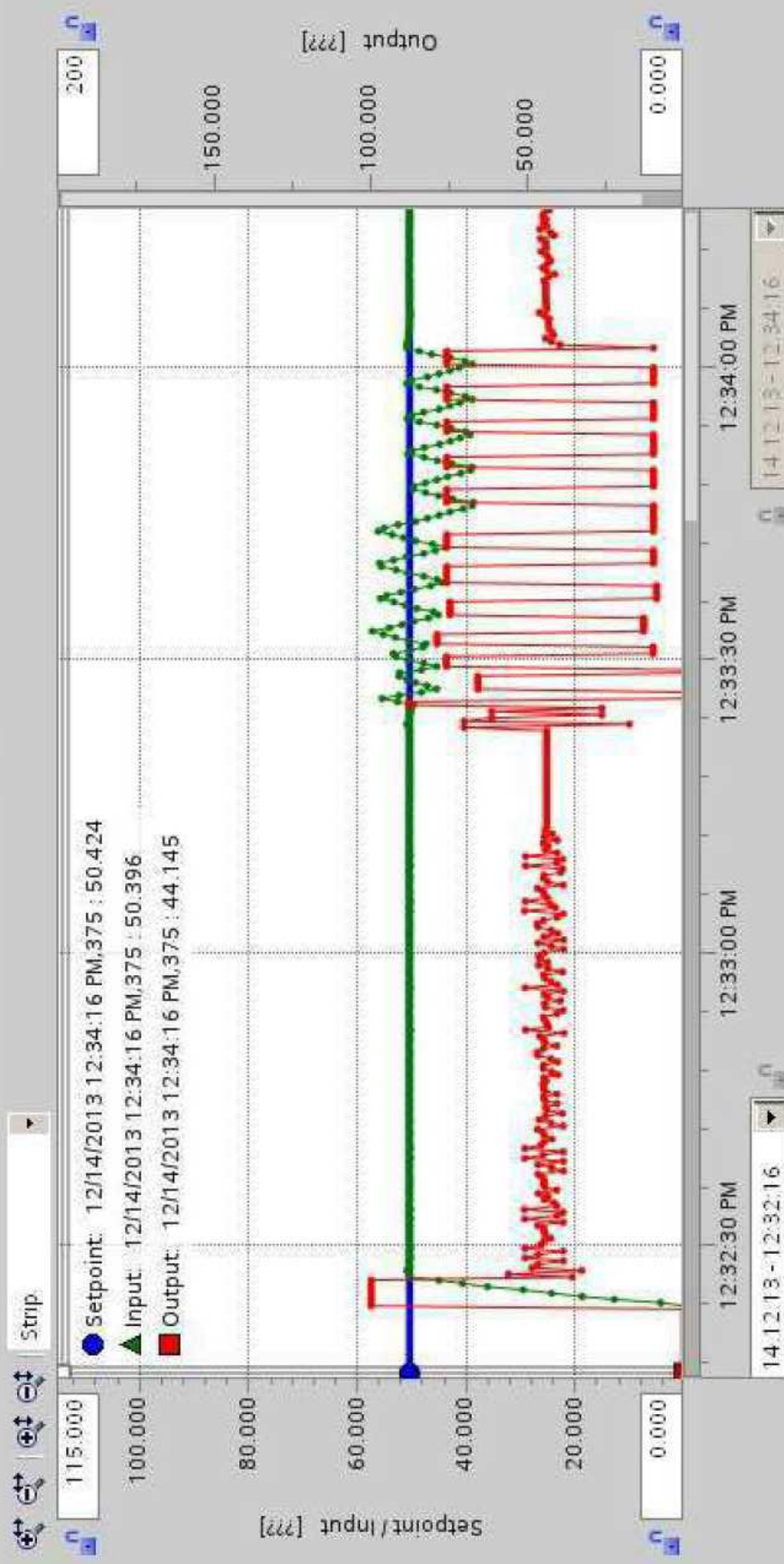
Stop

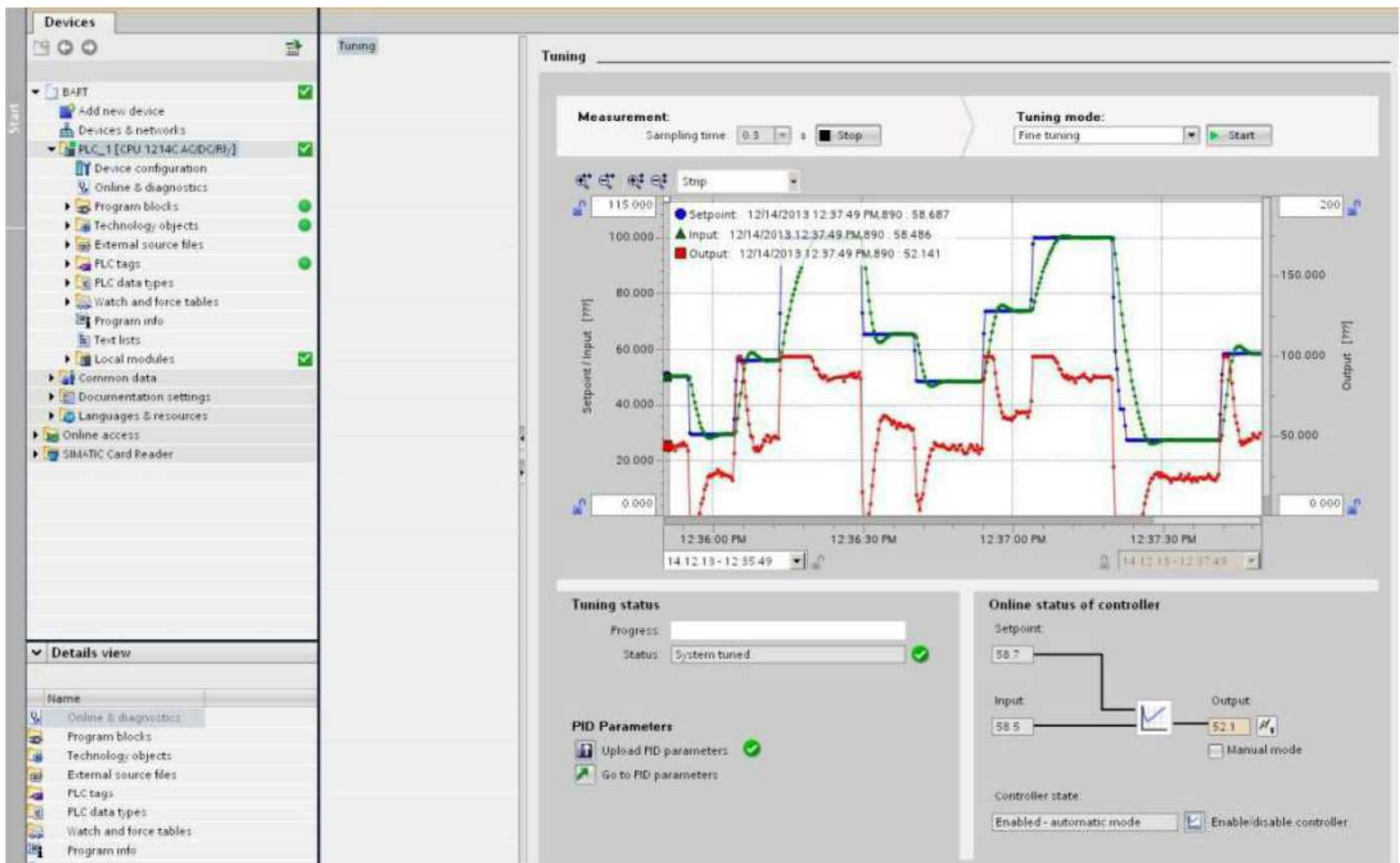
Tuning mode:

Fine tuning



Start



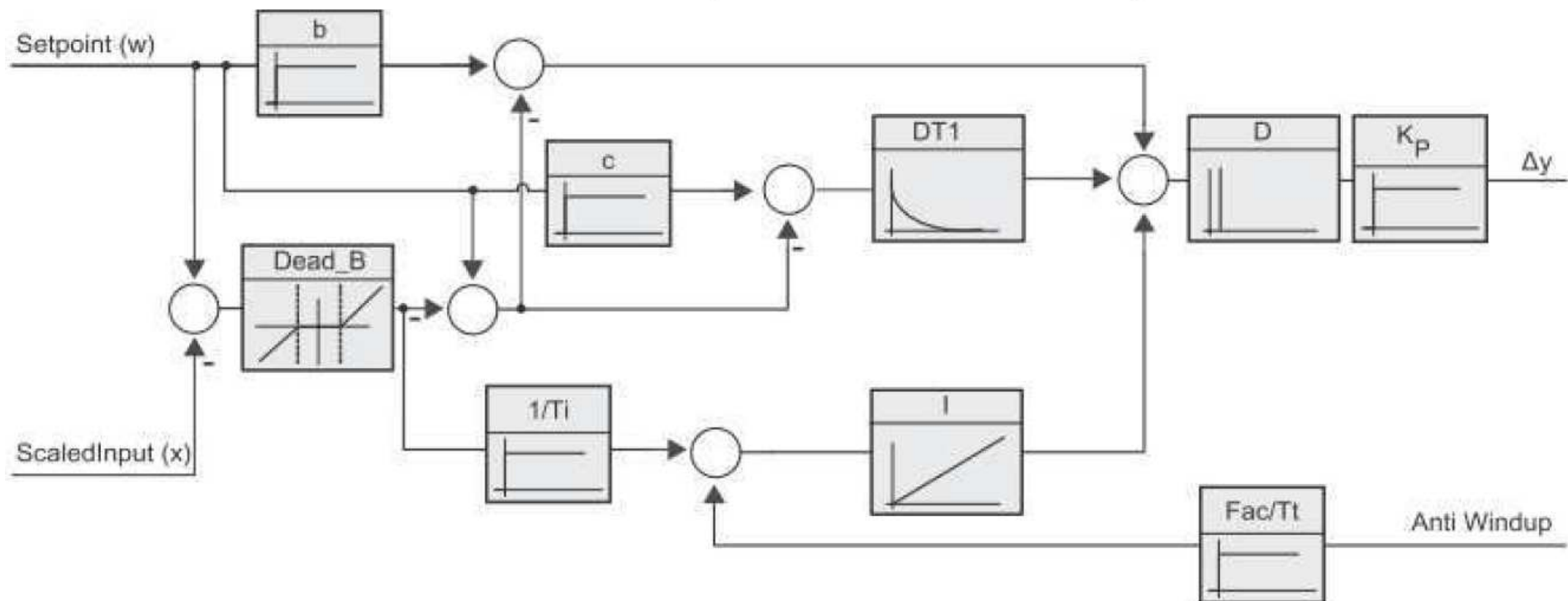


TIA Portal: PID Compact - (Auto-)Tuning a PID Controller!

<https://youtu.be/oWISLJKMQh8>

PID 3Step

$$\Delta y = K_p \cdot s \cdot \left[(b \cdot w - x) + \frac{1}{T_i \cdot s} (w - x) + \frac{T_D \cdot s}{a \cdot T_D \cdot s + 1} (c \cdot w - x) \right]$$



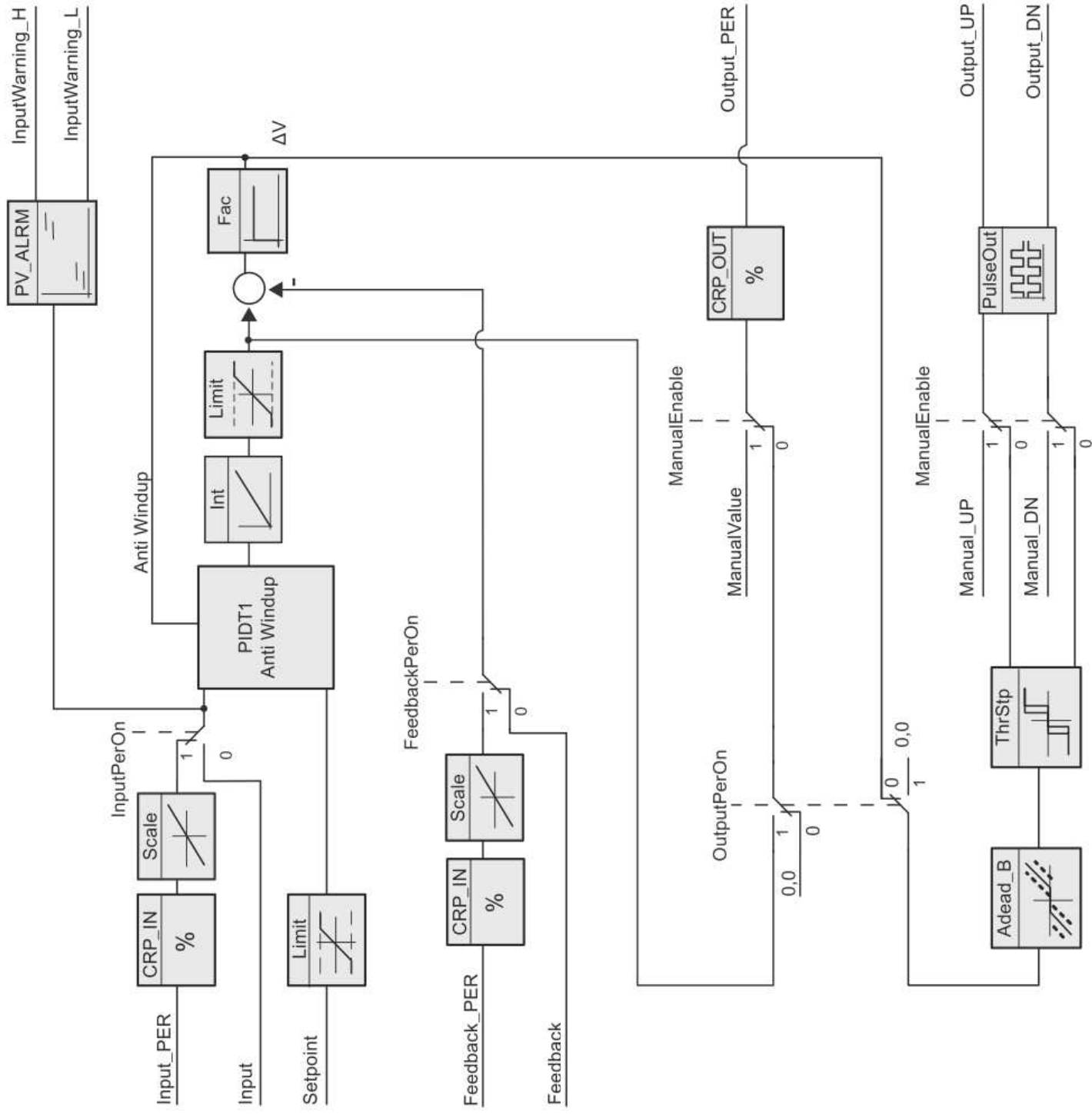


Figure 9-5 Operation of the PID_3Step controller the position feedback enabled

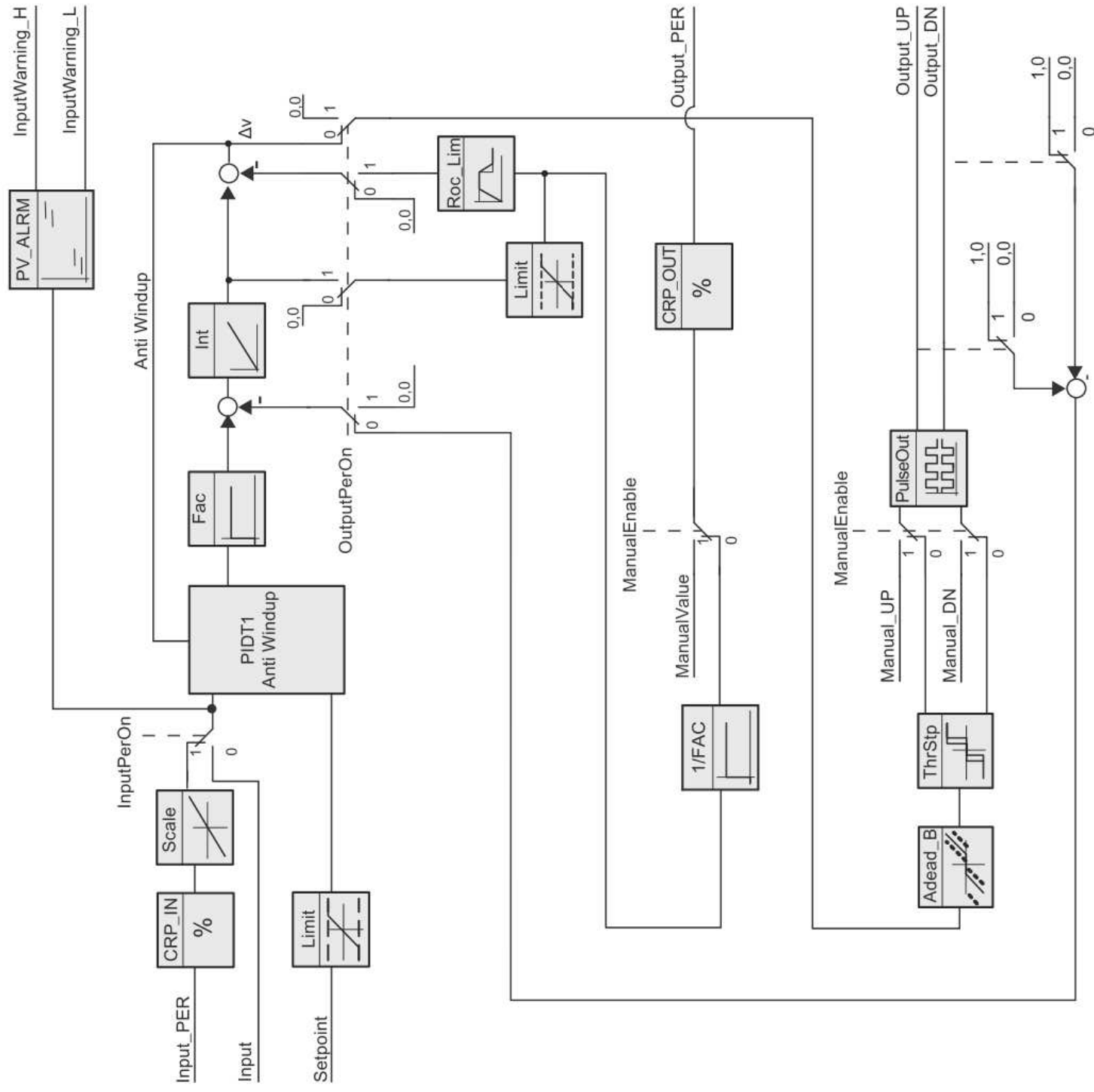


Figure 9-4 Operation of the PID_3Step controller without position feedback

Controller type

☐ General % ☐ Invert the control logic

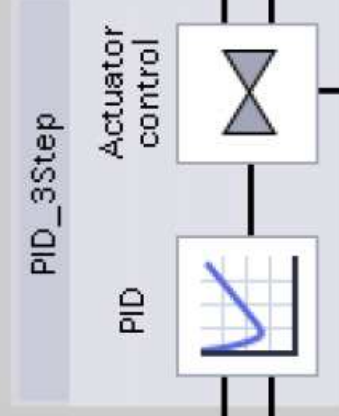
☒ Enable last mode after CPU restart.

Input / output parameters

Setpoint: %

Input: Input_PER (analog)

Output: Output (digital)



Feedback:

No Feedback (default)

☐ Actuator endstop

Actuator_H:

Actuator_L: